

WORK SHEET - 02

VECTOR & 3D

[SINGLE CORRECT CHOICE TYPE]

- Q.1 A normal line with positive direction cosines to the plane P makes equal angle with coordinate axes. The distance of the point A(1, 2, 3) from the line $\frac{x-1}{1} = \frac{y+2}{1} = \frac{z-3}{2}$ measured parallel to the plane P is equal to
(A) $\sqrt{14}$ (B) $\sqrt{30}$ (C) $\sqrt{35}$ (D) 7
- Q.2 $\vec{a}_1, \vec{a}_2, \vec{a}_3$ are non zero and non-coplanar vectors such that $(x + y - 3)\vec{a}_1 + (2x - y + 2)\vec{a}_2 + (2x + y + \lambda)\vec{a}_3 = \vec{0}$ holds for some x and y, then λ is
(A) $\frac{7}{3}$ (B) 2 (C) $-\frac{10}{3}$ (D) $\frac{5}{3}$
- Q.3 Equation of line in the plane $\Pi : 2x - y + z - 4 = 0$ which is perpendicular to the line l whose equation is $\frac{x-2}{1} = \frac{y-2}{-1} = \frac{z-3}{-2}$ and which passes through the point of intersection of l and Π , is
(A) $\frac{x-2}{3} = \frac{y-1}{5} = \frac{z-1}{-1}$ (B) $\frac{x-1}{3} = \frac{y-3}{5} = \frac{z-5}{-1}$
(C) $\frac{x+2}{2} = \frac{y+1}{-1} = \frac{z+1}{1}$ (D) $\frac{x-2}{2} = \frac{y-1}{-1} = \frac{z-1}{1}$
- Q.4 Equation of plane which passes through the point of intersection of lines $\frac{x-1}{3} = \frac{y-2}{1} = \frac{z-3}{2}$ and $\frac{x-3}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ and at greatest distance from the point (0, 0, 0) is
(A) $4x + 3y + 5z = 25$ (B) $4x + 3y + 5z = 50$
(C) $3x + 4y + 5z = 49$ (D) $x + 7y - 5z = 2$
- Q.5 If $\vec{V}_1 = \hat{i} + \hat{j} + \hat{k}$ and $\vec{V}_2 = a\hat{i} + b\hat{j} + c\hat{k}$ where $a, b, c \in \{-2, -1, 0, 1, 2\}$, then number of possible non-zero vectors $\vec{V}_2$ perpendicular to $\vec{V}_1$ is
(A) 10 (B) 13 (C) 15 (D) 18
- Q.6 Centroid of the tetrahedron OABC, where A $\equiv$ (a, 2, 3), B $\equiv$ (1, b, 2), C $\equiv$ (2, 1, c) and O is the origin is (1, 2, 3) then the value of $(a^2 + b^2 + c^2)$ is equal to
(A) 75 (B) 80 (C) 121 (D) 125

- Q.7 The direction ratios of a normal to the plane through $(1, 0, 0)$, $(0, 1, 0)$ which makes an angle of $\sec^{-1}\sqrt{2}$ with the plane $x + y = 3$ is
 (A) $(1, \sqrt{2}, 1)$ (B) $(1, 1, \sqrt{2})$ (C) $(1, 1, 2)$ (D) $(\sqrt{2}, 1, 1)$
- Q.8 If the angles between the vectors $\vec{v}_1$ and $\vec{v}_2$, $\vec{v}_2$ and $\vec{v}_3$, $\vec{v}_3$ and $\vec{v}_1$ be $\frac{\pi}{3}$, $\frac{\pi}{4}$ and $\frac{\pi}{6}$ respectively, then the angle the vector $\vec{v}_1$ makes with the plane containing $\vec{v}_2$ and $\vec{v}_3$ is
 (A) $\sin^{-1}\sqrt{1-\sqrt{\frac{2}{3}}}$ (B) $\sin^{-1}\sqrt{\sqrt{\frac{3}{2}}-1}$ (C) $\cos^{-1}\sqrt{1-\sqrt{\frac{2}{3}}}$ (D) $\cos^{-1}\sqrt{\sqrt{\frac{3}{2}}-1}$
- Q.9 Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$, $\vec{b} = \hat{i} + \hat{j}$. If $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} = |\vec{c}|$, $|\vec{c} - \vec{a}| = 2\sqrt{2}$ and the angle between $\vec{a} \times \vec{b}$ and $\vec{c}$ is 30° , then $|(\vec{a} \times \vec{b}) \times \vec{c}|$ equals
 (A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) 3 (D) $\frac{3\sqrt{3}}{2}$
- Q.10 The shortest distance between the lines $2x + y + z - 1 = 0 = 3x + y + 2z - 2$ and $x = y = z$, is
 (A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$ (C) $\frac{3}{\sqrt{2}}$ (D) $\frac{\sqrt{3}}{2}$

[PARAGRAPH TYPE]

Paragraph for question nos. 11 & 12

Let $\hat{x}, \hat{y}$ and $\hat{z}$ be unit vectors such that $\hat{x} + \hat{y} + \hat{z} = \vec{a}$, $\hat{x} \times (\hat{y} \times \hat{z}) = \vec{b}$, $(\hat{x} \times \hat{y}) \times \hat{z} = \vec{c}$, $\vec{a} \cdot \hat{x} = \frac{3}{2}$, $\vec{a} \cdot \hat{y} = \frac{7}{4}$ and $|\vec{a}| = 2$.

- Q.11 The value of $\hat{y} \cdot \hat{z}$ is equal to
 (A) $\frac{-1}{4}$ (B) $\frac{3}{4}$ (C) 0 (D) $\frac{1}{2}$
- Q.12 The vector $\hat{x}$ in terms of $\vec{a}, \vec{b}$ and $\vec{c}$ is equal to
 (A) $\frac{2}{3}(3\vec{a} + 4\vec{b} + 8\vec{c})$ (B) $\frac{1}{3}(3\vec{a} + 4\vec{b} + 8\vec{c})$
 (C) $\frac{2}{3}(3\vec{a} + 2\vec{b} + 4\vec{c})$ (D) $\frac{1}{3}(3\vec{a} + 2\vec{b} + 4\vec{c})$

Paragraph for question nos. 13& 14

Consider, three vectors $\vec{v}_1 = \sin \theta \hat{i} + \cos \theta \hat{j} + (a - 3)\hat{k}$

$$\vec{v}_2 = (\sin \theta + \cos \theta)\hat{i} + (\cos \theta - \sin \theta)\hat{j} + (b - 4)\hat{k}$$

$$\vec{v}_3 = \cos \theta \hat{i} - \sin \theta \hat{j} + (c - 5)\hat{k}$$

- Q.13 If $\vec{v}_1 + \vec{v}_2 + \vec{v}_3 = \lambda \hat{i}$ where $\theta \in [0, 2\pi]$ and $a, b, c \in \mathbb{N}$ then number of quadruplets (a, b, c, θ) are
(A) 55 (B) 91 (C) 110 (D) 182

- Q.14 If $\vec{v}_1, \vec{v}_2, \vec{v}_3$ represent sides of a triangle PQR in xy-plane and corresponding to triangle PQR there exist one more triangle ABC whose sides are denoted by a, b, c then $\frac{\text{Ar}(\triangle ABC)}{\text{Ar}(\triangle PQR)}$ is equal to
(A) 6 (B) 9 (C) 12 (D) 24

**[MULTIPLE CORRECT CHOICE TYPE]
(ONE OR MORE THAN ONE)**

- Q.15 Consider the lines $L_1: \vec{r} = \hat{i} - 3\hat{k} + \lambda(2\hat{i} - \hat{j} + \hat{k})$, $L_2: \vec{r} = 4\hat{i} - 3\hat{j} - 3\hat{k} + \mu(\hat{i} + \hat{j} + 2\hat{k})$, ($\lambda, \mu \in \mathbb{R}$) and the planes $\sigma_1: \vec{r} \cdot (7\hat{i} + \hat{j} + 2\hat{k}) - 3 = 0$, $\sigma_2: \vec{r} \cdot (3\hat{i} + 5\hat{j} - 6\hat{k}) = 4$.

Let $ax - 3y + bz = c$ be the equation of plane passing through the point of intersection of lines L_1 and L_2 , and perpendicular to the planes σ_1 and σ_2 . Then

- (A) $a + b = -3$ (B) $a + b = -1$ (C) $b + c = 12$ (D) $b + c = 11$

- Q.16 Let $A(1, 1, 1)$, $B(2, 3, 5)$, $C(-1, 0, 2)$ be three points, then equation of a plane parallel to the plane ABC which is at a distance 2 from it, is

- (A) $2x - 3y + z + 2\sqrt{14} = 0$ (B) $2x - 3y + z - 2\sqrt{14} = 0$
(C) $3x - 3y + z + 2 = 0$ (D) $2x - 3y + z - 2 = 0$

- Q.17 Let $L_1: \frac{x-1}{2} = \frac{y-2}{-1} = \frac{z-3}{3}$, $L_2: \frac{x-1}{-1} = \frac{y-2}{3} = \frac{3(z-3)}{5}$ and $L_3: \frac{x-1}{-32} = \frac{y-2}{-19} = \frac{z-3}{15}$

be three lines. A plane intersecting these lines at A, B and C respectively such that $PA = 2$, $PB = 3$ and $PC = 6$ where $P \equiv (1, 2, 3)$. If V is the volume of the tetrahedron PABC and d is the perpendicular distance of the plane from the point P then-

- (A) $V = 18$ cubic units (B) $V = 6$ cubic units
(C) $d = \frac{6}{\sqrt{14}}$ units (D) $d = 7$ units

- Q.18 If $\vec{a}, \vec{b}$ are non-zero and non-collinear vectors, then $\vec{v} = \frac{\vec{a}}{|\vec{a}|} + \frac{\vec{b}}{|\vec{b}|}$ is

- (A) a unit vector (B) in the plane of $\vec{a}$ and $\vec{b}$.
(C) equally inclined to $\vec{a}$ and $\vec{b}$ (D) perpendicular to $\vec{a} \times \vec{b}$

[INTEGER TYPE]

Q.19 If vectors $\vec{OA}$, $\vec{OB}$ and $\vec{OC}$ are coplanar such that $\vec{OA} \wedge \vec{OB} = \vec{OB} \wedge \vec{OC} = 30^\circ$ and area of parallelograms OADB, OBEC and ΔOAC are 6, 10 and 4 square units respectively. If ODFE is a parallelogram and area of quadrilateral BDFE is S, then find the value of $\left\lfloor \frac{S}{3} \right\rfloor$.

[Note : $\lfloor k \rfloor$ denotes greatest integer less than or equal to k.]

Q.20 Line L meets lines $L_1 : \frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ and $L_2 : \frac{x-2}{2} = \frac{y-1}{4} = \frac{z-4}{5}$ orthogonally at points P and Q.

$(PQ)^2$ is D. DRs of line L are (a, b, c) $\{a, b, c \in I\}$, then least value of $\frac{5D}{3} + |a| + |b| + |c|$.

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ANSWER KEY

Q.1	A	Q.2	C	Q.3	B	Q.4	B	Q.5	D
Q.6	A	Q.7	B	Q.8	B	Q.9	B	Q.10	A
Q.11	C	Q.12	B	Q.13	C	Q.14	C	Q.15	BD
Q.16	AB	Q.17	BC	Q.18	BCD	Q.19	[5]	Q.20	[6]

